
VitaCheck: Symptom-Based Vitamin Deficiency Detection and Personalized Meal Planning

Afeef Ali, Sanjana Gollu, Aarav Bafna, Natalie Rogers, Tiffany Widjaja
Virginia Tech
aali11@vt.edu, sanjanag@vt.edu, aaravb@vt.edu
natalier06@vt.edu, tiffanywid@vt.edu

1 Introduction

Vitamin deficiencies are some of the most prevalent yet underdiagnosed conditions globally. Symptoms such as fatigue, hair loss, and muscle weakness are often dismissed, whereas blood tests are inaccessible or inconvenient for many patients. Early identification of deficiencies like anemia, scurvy, or rickets can prevent serious long-term health consequences, yet no accessible tool currently allows a person to describe their symptoms in plain language and receive both a likely diagnosis and a dietary intervention in one interaction.

This proposal presents the plan for a capstone project that addresses this gap directly. We develop a machine learning pipeline that classifies vitamin deficiency type from patient-reported symptoms and demographics, then passes the predicted diagnosis to a large language model to generate a personalized meal plan. This system will provide an easily accessible screening tool that could help individuals make informed dietary decisions.

What makes our approach unique is the combination of three elements that prior work has never combined: multi-class symptom-based deficiency classification without blood work, free-text natural language input parsed by an LLM, and an automatically generated dietary intervention tied to the predicted diagnosis. We review relevant literature below to establish where existing methods succeed, where they fall short, and how our approach builds on and extends them.

2 Changes Since Proposal Submission

The following changes have been made in response to professor feedback on the graded proposal:

- **Class imbalance identified and addressed:** As suggested, we examined the class distribution and confirmed the dataset is significantly unbalanced. Healthy and Anemia together account for nearly 70% of records while Scurvy (95 records, 2.4%) and Night_Blindness (122 records, 3.1%) are severely underrepresented. We adopted macro-averaged F1 as our primary metric rather than accuracy, which penalizes the model equally for missing any class regardless of size. On the LLM side, we plan to address imbalance by prompting the LLM to explicitly consider all five disease classes when parsing symptoms rather than defaulting to the most common ones.
- **Project title added:** The project is now titled *VitaCheck: Symptom-Based Vitamin Deficiency Detection and Personalized Meal Planning*.
- **Data leakage identified and resolved:** During initial model training we discovered that clinical lab value columns and RDA intake columns directly encode the diagnosis. Training with these columns produced 99%+ accuracy but was clinically meaningless — a real user would not have access to these values. We removed all lab and RDA columns and retrained on self-reportable features only, producing an honest accuracy of 84% and macro F1 of 0.69.

- **All four models trained and compared:** Logistic Regression, Random Forest, LightGBM, and XGBoost are all now trained and evaluated with accuracy and macro F1 reported for each.
- **Evaluation metrics expanded:** The notebook now reports macro F1, per-class precision/recall, a confusion matrix, and 5-fold cross-validation.

3 Related Work Survey

We reviewed multiple prior works relevant to our project domain. These papers represent different algorithmic approaches, theoretical perspectives, and evaluation strategies.

3.1 Large language models in clinical nutrition: an overview of its applications, capabilities, limitations, and potential future prospects (Belkhouribchia & Pen, 2025)

This paper examines a number of studies related to using LLMs in clinical nutrition and summarizes the findings and limitations. Importantly to our project, it covers many examples regarding the creation of food plans or personalized diets. This helps us to understand what methods do and don't produce accurate, readable outcomes. Even more specifically, there are a couple that focus on creating food plans for particular diagnoses, such as malnutrition, diabetes, or obesity. These provide information about methods for developing food plans centered around a diagnosis, like vitamin deficiencies. What these seem to lack, however, is allowing the patient to directly interface with the LLM to obtain a diagnosis, followed by the development of a personalized diet.

This paper really delves into the strengths and weaknesses of the use of LLMs and where they excel and fall short in being used for clinical nutrition. As such, a main strength is that it allows us to identify ways to improve the functionality of the LLM and understand some of the limitations. On the other hand, it does not cover our main problem of diagnosing vitamin deficiencies from provided symptoms.

3.2 Machine learning-based prediction of vitamin D deficiency: NHANES 2001–2018 (Guo, He, & Li, 2024)

This study uses six ML models (XGBoost, random forest, logistic regression, SVM, neural network, and GBM) to predict vitamin D deficiency from just nine easily collected interview variables—things like age, race, BMI, milk consumption, and diabetes status—across 62,919 NHANES participants. This is relevant to our project because it demonstrates the feasibility of predicting a vitamin deficiency without requiring any blood work, using only demographic and lifestyle factors that a patient could self-report. The paper also deploys a practical Shiny web calculator and uses SHAP values to explain which features drive predictions, which aligns with the goal of building an accessible, interpretable tool.

A major strength is the massive, nationally representative dataset and the comprehensive metric reporting, including calibration (Brier score) and clinical utility (decision curve analysis). However, there is a critical weakness: the best model (XGBoost) reports a perfect AUC of 1.0 from nine simple interview questions, which is biologically implausible and strongly suggests data leakage or overfitting. Additionally, the study uses only a random train/test split with no external or temporal validation, race is the top predictor which raises fairness concerns, and season of blood draw—a known dominant driver of vitamin D levels—is entirely absent from the feature set.

3.3 Machine Learning-Driven Prediction of Vitamin D Deficiency Severity with Hybrid Optimization (Bhimavarapu, Battineni, & Chintalapudi, 2025)

This paper is quite similar to the previous one by Guo, He, and Li. Where it differs, however, is that this paper is focused on classifying the severity of the vitamin D deficiency (VDD). Based on 12 parameters collected from questionnaires, hospital records, and medical imaging, a dataset was preprocessed, and then the improved whale optimization (IWOA) algorithm was used to select 12 features for the severity prediction. Using 6 different machine learning models, the study evaluated key performance metrics such as precision, recall, F1-score, and accuracy. Ultimately, the models performed impressively well, with accuracy of up to 99.4%, as well as high precision, recall, and

F1-scores. This suggests a strong case for using ML to accurately predict VDD severity. While our project is focused on the diagnoses and not necessarily the classification, it's still incredibly helpful to understand the methods used and the strengths and weaknesses of each. It's also encouraging to see that ML methods can very accurately predict medical conditions based on patient data.

3.4 Efficient Prediction of Vitamin B Deficiencies via Machine-Learning Using Routine Blood Test Results in Patients With Intense Psychiatric Episode (Tamune et al., 2020)

In this paper, they train random forest, SVM, logistic regression, and KNN models to predict vitamin B1, B12, and folate deficiency from 31 routine blood test parameters in 497 psychiatric emergency patients. This is important to our project because it is one of the very few papers that directly tackle B-vitamin deficiency prediction, and it honestly reveals a key challenge: folate deficiency was reasonably predictable (AUC 0.80), but B12 deficiency prediction was essentially no better than chance (AUC 0.60). This tells us that B12 deficiency may require different feature types—such as dietary history, medication use, or functional biomarkers like methylmalonic acid—rather than standard CBC parameters alone.

A notable strength is that it is the only paper in the broader literature that uses a temporal validation split (training on earlier months, testing on later months), which gives more realistic performance estimates than random splitting. It also tests multiple deficiency thresholds for robustness and provides interpretable feature importance. The main limitations are the very small sample size (only 13 B12-deficient patients in the validation set), the single-center all-Asian psychiatric population, which limits generalizability, and the fact that it does not allow a patient to interface directly with the model to obtain a diagnosis or personalized intervention plan.

3.5 Predictors of Micronutrient Deficiency Among Children Aged 6–23 Months in Ethiopia: A Machine Learning Approach (Gebeye LG, Dessie EY, & Yimam JA, 2024)

The paper applies five supervised ML models (Random Forest, SVM, Logistic Regression, Neural Network, and Naïve Bayes) to nationally representative 2019 EMDHS data to predict micronutrient deficiency using demographic and socioeconomic variables rather than lab biomarkers. Random Forest performed best (AUROC 80.01 percent, accuracy 72.41 percent), identifying region, wealth index, maternal education, media exposure, and child age as key predictors. A major strength is the use of a large population level dataset and comparison across multiple models, along with spatial mapping and logistic regression benchmarking for interpretability. However, the outcome is based on a 24-hour dietary recall rather than biochemical confirmation, which weakens clinical validity, and the cross-sectional design limits causal inference. While performance is reasonable, it is not strong enough for diagnostic deployment, and deeper explainability methods are absent. Overall, the study supports the feasibility of non-invasive ML-based deficiency prediction but highlights the need for improved outcome measurement and validation.

3.6 Differential Diagnosis of Iron Deficiency Anemia from Aplastic Anemia Using Machine Learning and Explainable AI Utilizing Blood Attributes (Darshan BSD, Sampathila N, Bairy GM, Prabhu S, Belurkar S, Chadaga K, & Nandish S, 2025)

The paper develops ML models to distinguish iron deficiency anemia (IDA) from aplastic anemia (AA) using 23 routine blood parameters from a 500-patient retrospective dataset. Multiple classifiers and ensemble models were evaluated, with LightGBM and stacked models achieving high accuracy (about 96 percent), and five explainable AI techniques (SHAP, LIME, Eli5, Qlattice, Anchor) were used to interpret predictions. A major strength is its strong emphasis on explainability and clinical interpretability, making the model more trustworthy for decision support. It also demonstrates how deficiency-related conditions can be predicted from standard laboratory inputs. However, the study is limited by its small, single-center dataset and lack of external validation, which raises concerns about generalization. Unlike survey-based or symptom-based models in other papers, this study relies entirely on blood biomarkers, making it more clinically realistic but less applicable for early, non-invasive screening. Together with the other papers, it adds a strong explainable AI and differential diagnosis dimension while highlighting the need for broader validation and integration with earlier-stage diagnostic approaches.

3.7 Strengths and Weaknesses of Prior Work

The four papers share a consistent pattern, which is that existing ML approaches to vitamin deficiency prediction are either narrowly focused on a single vitamin, require clinical data like blood draws that patients cannot self-report (Tamune et al.), or evaluate prediction without connecting the diagnosis to any actionable intervention. The two vitamin D studies (Guo et al.; Bhimavarapu et al.) demonstrate that high classification accuracy is achievable from self-reported data alone, but neither allows a patient to interface directly with the model using free-text symptom descriptions. The LLM survey (Belkhouribchia & Pen) shows that language models can generate personalized dietary plans from a known diagnosis, but no reviewed work combines symptom-based diagnosis with LLM-driven meal planning in a single end-to-end pipeline.

Our work directly addresses these gaps in three ways. First, we extend the scope from single-vitamin prediction to multi-class classification across six deficiency categories simultaneously. Second, we replace structured clinical inputs with free-text symptom descriptions parsed by an LLM, making the tool accessible without any medical testing. Third, we close the loop that prior work leaves open by feeding the ML prediction directly into an LLM to generate a personalized meal plan, something none of the reviewed papers attempted.

4 Project Setup

To set up our project, we began with sharing a GitHub with all team members so that we can all access the same data and code. We also set-up a Google Colab where all the code will be run and will be updated in real time.

5 Data Collection Process

We will use datasets with clearly specified characteristics:

- CSV structured format
- About 4k records
- 35+ variable features
- Training/validation/test splits

Labeled data from public datasets to feed into a model for training purposes. This will be formatted in a .csv sheet style with columns of data. There will be categorical and numerical data since there will be symptoms that the user is required to enter. These symptoms will be given to the machine learning model. Data will be obtained from public datasets like Kaggle and be formatted by numerical (like age and percentage) and categorical (a minimum of 10 columns of symptoms).

5.1 Dataset Source and Selection

Our primary dataset is the *Vitamin Deficiency Disease Prediction Dataset* sourced from Kaggle (uploaded January 2026), containing 4,000 patient records across 34 columns. The dataset includes demographic information, lifestyle factors, clinical lab values, and binary symptom indicators, with `disease_diagnosis` as the target variable. Five disease classes are present: **Healthy** (1,509 records, 37.7%), **Anemia** (1,245 records, 31.1%), **Rickets_Osteomalacia** (1,029 records, 25.7%), **Night_Blindness** (122 records, 3.1%), and **Scurvy** (95 records, 2.4%). The class distribution is notably imbalanced: Healthy and Anemia together account for nearly 70% of records, while Night Blindness and Scurvy together represent fewer than 6%. This imbalance makes macro-averaged F1 the appropriate primary metric, as it penalizes models that ignore minority classes.

5.2 Feature Schema

The 34 features fall into four categories. **Demographic and lifestyle features** (8 columns) include age (18–84 years), gender (Male/Female), BMI (15.0–45.0), smoking status (Never/Former/Current), alcohol consumption (Moderate/Heavy), exercise level (Sedentary/Light/Moderate/Active), diet

type (Omnivore/Vegetarian/Vegan/Pescatarian), sun exposure (Low/Moderate/High), income level (Low/Middle/High), and latitude region (Low/Mid/High).

Nutritional intake features (8 columns) report each patient’s intake of vitamins A, C, D, E, B12, folate, calcium, and iron as a percentage of the recommended daily allowance (% RDA). Mean intake levels across the dataset range from 62.7% RDA (vitamin B12) to 90.8% RDA (vitamin A), indicating widespread sub-threshold intake, particularly for B12 and vitamin D (mean 72.2% RDA).

Clinical lab values (4 columns) include hemoglobin (g/dL), serum vitamin D (ng/mL), serum vitamin B12 (pg/mL), and serum folate (ng/mL).

Symptom features (8 binary columns) indicate the presence or absence of: night blindness, fatigue, bleeding gums, bone pain, muscle weakness, numbness/tingling, memory problems, and pale skin. A `symptoms_count` column aggregates the total number of active symptoms per record (range 0–9, mean 1.87).

5.3 Initial Findings and Summary Statistics

Table 1: Class Distribution

Class	Count	Percentage
Healthy	1509	37.7%
Anemia	1245	31.1%
Rickets_Osteomalacia	1029	25.7%
Night_Blindness	122	3.1%
Scurvy	95	2.4%

Per-class symptom analysis reveals strong class-specific signals in several features. Bleeding gums is present in 73.7% of Scurvy patients but fewer than 4% of all other classes, making it the single most discriminative symptom in the dataset. Night blindness appears in 81.1% of Night_Blindness patients versus 3.0% of Healthy patients. Bone pain is present in 68.3% of Rickets_Osteomalacia patients compared to 11.8% of Healthy patients. Fatigue, muscle weakness, and numbness/tingling are broadly distributed across Anemia, Rickets, and Scurvy, making them individually low-discriminative but potentially useful in combination. The Healthy class has near-zero prevalence across all symptom features, which should make it the easiest class to classify correctly.

Two columns contain missing values: `alcohol_consumption` (1,278 missing, 32.0%) and `symptoms_list` (1,328 missing, 33.2%). The `symptoms_list` column is a free-text string redundant with the binary symptom indicators and will be dropped. `alcohol_consumption` will be imputed with a “Unknown” category rather than dropped, as missingness may itself be informative.

5.4 Data Preprocessing Plan

The following preprocessing steps will be applied prior to model training: the `symptoms_list` free-text column will be dropped as it is superseded by the binary indicators; `alcohol_consumption` missing values will be filled with an “Unknown” category; label encoding will be applied to the target variable `disease_diagnosis`; one-hot encoding will be applied to all multi-category categorical features (diet type, exercise level, etc.); standard scaling (zero mean, unit variance) will be applied to all continuous numerical features (age, BMI, % RDA columns, lab values); and a stratified 80/20 train/test split with 5-fold cross-validation will be used to preserve class proportions across all splits, which is critical given the significant imbalance between the majority classes (Healthy, Anemia) and minority classes (Night_Blindness, Scurvy).

During implementation we found that the clinical lab value columns and all RDA intake columns directly encode the diagnosis and were removed to prevent data leakage. The final model trains on demographic, lifestyle, and binary symptom features only, as these are the only values a real user could self-report.

6 Mathematical Background

6.1 Multi-Class Classification

Our task requires assigning one of five disease labels to each patient based on self-reported symptoms and demographics. The model learns from labeled examples and outputs the most likely disease class for a new input.

6.2 Gradient Boosted Trees

XGBoost and LightGBM build many simple decision trees sequentially, where each new tree corrects the mistakes of the previous ones. A learning rate controls how much each tree contributes, and after hundreds of trees are combined the final prediction is their accumulated result. LightGBM uses a faster tree-building strategy that reduces training time while achieving comparable accuracy.

6.3 Baseline Models

Logistic Regression learns a linear relationship between features and classes and serves as our lower bound. Random Forest builds many independent trees in parallel and takes a majority vote, serving as a strong non-linear baseline.

6.4 Evaluation Metrics

Plain accuracy is misleading on our imbalanced dataset — a model that always predicts Healthy would score 37.7% while ignoring four disease classes entirely. We use macro-averaged F1 as our primary metric, which scores each class separately and averages them equally regardless of class size, penalizing the model for ignoring rare classes like Scurvy and Night Blindness.

7 Model Approach

We used XGBoost and LightGBM as our primary models to predict vitamin deficiency from symptoms, demographics, and lifestyle factors. We chose these because they typically give higher accuracy, train faster, and handle overfitting better than simpler models on health data like this. Random Forest and Logistic Regression serve as our baselines, with Random Forest providing a strong comparison point and Logistic Regression acting as a minimal floor so we can clearly measure the improvement our primary models achieve. We plan on transforming our input to make sure that our model will get a high accuracy and make it easier for the model to find patterns. This includes increasing the number of specific columns using given data, adding more weights for certain parameters, and fine-tuning the model to prevent overfitting to get the highest accuracy.

Free-text user input will be parsed into structured features using an open-source LLM (Qwen3.5-14B-Instruct, but may change), and the prediction result will then be passed to the same LLM to generate a personalized meal plan.

```
# -----  
# 3. Train/test split  
# -----  
X_train, X_test, y_train, y_test = train_test_split(  
    X, y_encoded, test_size=0.2, random_state=42, stratify=y_encoded  
)  
  
# 5. LightGBM  
# -----  
lgb_model = lgb.LGBMClassifier(n_estimators=500, learning_rate=0.05, num_leaves=31, random_state=42)  
lgb_model.fit(X_train, y_train)  
print("LightGBM Accuracy:", accuracy_score(y_test, lgb_model.predict(X_test)))
```

Figure 1: LightGBM model training code showing hyperparameters used: 500 estimators, learning rate of 0.05, and 31 leaves.

Potential challenges: The primary difficulty encountered was data leakage from clinical lab values and RDA columns, which inflated accuracy to 99%+ but made the model useless in practice. Resolving this required removing 12 columns and reframing the problem around what a real user could self-report. Class imbalance remains the central ongoing difficulty, with only 95 Scurvy and 122 Night Blindness records, the model has limited signal for these classes. The LLM parsing step also

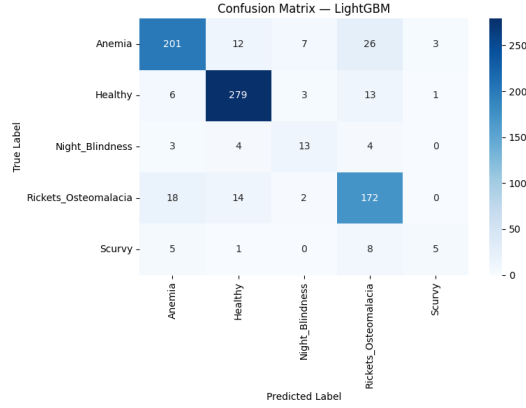


Figure 2: Confusion matrix for LightGBM trained on self-reportable features only. The model performs well on majority classes but struggles with minority classes Scurvy and Night_Blindness due to limited training examples (95 and 122 records respectively).

introduces an additional error source not present in structured-input models, as the LLM must correctly map informal language like “feeling drained” to the binary symptom features the classifier expects.

7.1 Training Milestones

Model training will proceed in phases aligned with the course timeline. In the first phase, Logistic Regression and Random Forest baselines are trained on the full feature set to establish a performance floor and ceiling. In the second phase, XGBoost and LightGBM are trained with default hyperparameters to confirm they outperform the baselines. In the third phase, hyperparameter tuning via grid search with stratified 5-fold cross-validation is applied to both primary models, targeting a macro F1 of at least 0.90 across all six classes.

7.2 Planned Feature Engineering

Prior to finalizing the feature set, we will remove symptoms with near-zero variance across the dataset, apply correlation filtering to drop features with pairwise correlation above 0.8, and examine SHAP global importance plots to identify and potentially remove features with negligible contribution to any class. We will also explore creating composite features, for example combining multiple co-occurring symptoms that tend to appear together for a specific deficiency.

7.3 Model Tuning Strategy

In addition to feature engineering, we perform targeted hyperparameter tuning to improve model performance and generalization. For gradient boosting models such as XGBoost and LightGBM, key hyperparameters include learning rate, maximum tree depth, and number of estimators. These parameters significantly impact model complexity and the ability to capture nonlinear relationships in the data.

Rather than increasing training iterations arbitrarily, we use grid search combined with stratified 5-fold cross-validation to systematically evaluate different parameter configurations. This approach ensures that performance improvements are robust across all classes, particularly given the class imbalance present in our dataset.

We also monitor for overfitting by comparing training and validation performance. Given the relatively small dataset size, controlling model complexity is critical to ensure that the model generalizes well to unseen data.

8 Evaluation Plan

Our main metric is macro-averaged F1, which weighs all 6 deficiency classes equally and handles class imbalance better than plain accuracy. We also report per-class precision and recall to see where the model struggles, ROC-AUC for overall performance, and a confusion matrix to understand which diseases get mixed up. We use a stratified 80/20 train-test split with 5-fold cross-validation so every class is represented proportionally in each fold. SHAP values tell us which symptoms matter most for each prediction.

8.1 Success Criteria

We consider the project successful if XGBoost or LightGBM achieves the highest macro F1 among all four models with consistent performance across all 5 cross-validation folds, demonstrating that gradient boosting methods outperform the baselines on this task.

9 Ongoing Work

Initial model training on self-reportable features only (no lab values) produced the following results:

Table 2: Model Performance — Symptoms and Demographics Only

Model	Accuracy	Macro F1
Logistic Regression	0.7800	0.5938
Random Forest	0.8350	0.6669
LightGBM	0.8375	0.6871
XGBoost	0.8337	0.6884

Gradient boosting models outperform both baselines as expected. Per-class results show strong performance on Healthy (F1 0.91) and Anemia (F1 0.83) but weaker performance on Night_Blindness (F1 0.53) and Scurvy (F1 0.36) due to limited training examples. 5-fold cross-validation on LightGBM confirmed consistent performance with a mean macro F1 of 0.677 and standard deviation of 0.027.

The following components are planned for the final version:

- **Feature engineering:** Interaction features, correlation filtering, and SHAP-based feature selection to improve minority class performance.
- **Hyperparameter tuning:** Grid search with 5-fold cross-validation on XGBoost and LightGBM.
- **SHAP explainability:** Visualizations showing which symptoms drive predictions per disease class.
- **LLM symptom parsing:** Converting free-text user input into the structured feature dictionary the model expects.
- **LLM meal plan generation:** Generating a personalized 7-day meal plan from the predicted diagnosis.
- **End-to-end pipeline:** Connecting all components so a user types symptoms in plain English and receives a diagnosis and meal plan in one interaction.

10 Conclusion

At this stage of the project we have established a working classification pipeline, identified and resolved a data leakage issue that affected early results, and produced honest baseline performance metrics across four models. The core challenge going forward is improving performance on the minority classes Scurvy and Night Blindness, which remain difficult to predict reliably given their limited representation in the dataset. Once the model is further refined, integrating the LLM pipeline for free-text symptom input and meal plan generation will complete the end-to-end system we set out to build. We believe the combination of an accessible symptom-based classifier and a personalized

dietary recommendation is a genuinely useful tool, and the results so far give us confidence that it is achievable.

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